TEXORA · LECTURE SHEET

Twine, Rope and Cable Manufacturing

A rope is a yarn built up in stages: fibres are twisted into yarns, yarns into strands, and strands are laid or braided into rope. This sheet covers the constructions, the direction conventions, the machinery in outline, and what the structure does to strength.

YM-4203 · TECHNICAL FIBRE & YARN · TOPIC 4 · LEVEL 4 TERM 2 · YARN MANUFACTURING SPECIALIZATION

1. Definitions and the construction hierarchy

- **Twine** — a light, strong cord made of two or more yarns twisted together, used for tying, packaging, agriculture (baler twine) and fishing. It is the smallest member of the family: essentially a plied yarn sold by length and strength rather than by diameter.
- **Rope** — a flexible, load-bearing structure, conventionally above about 4 mm diameter, built from strands laid, plaited or braided together, or from a braided cover over a core. A rope is a finished engineering product with a rated minimum breaking load.
- **Cable** — in the textile sense, a large rope, often for marine or towing duty; in the wider engineering sense the word also covers steel **wire rope** (strands of steel wires laid around a core) and electrical cables. Context decides which is meant — in a fibre-rope mill, “cable-laid” also names a construction in which already-laid ropes are laid together again.

Every laid rope follows the same hierarchy, and each level adds twist in a planned direction:

Fibre → Yarn → Strand → Rope (→ Cable-laid rope)

At each stage, many elements are combined so that load is shared, a single broken element does not fail the whole structure, and the assembly is flexible enough to bend over sheaves and be coiled, knotted and spliced.

2. Laid (twisted) ropes

2.1 The three-strand hawser lay

The classic construction is the **three-strand hawser-laid rope**. Its manufacture runs:

1. **Yarn formation** — fibres or filaments are twisted into rope yarns, historically on long rope walks, today on yarn-forming machines.

2. **Strand formation** — a defined number of yarns is twisted together, in the opposite direction to the yarn twist, to form a strand. Opposite directions at successive levels are what make the structure **torque-balanced enough** to stay closed rather than untwisting itself.
3. **Closing (laying)** — three strands are twisted together around their common axis, again reversing direction, to close the rope. The closing machine rotates the strand packages (or the take-up) while the strands are drawn together through a forming die that sets the lay.
4. **Finishing** — the rope is measured, sometimes heat-set or treated (for example against rot or for abrasion resistance), coiled or reeled, and its ends are whipped or fused to prevent unlaying.

2.2 Direction conventions: S and Z

Twist direction is named by the letter the central part of the helix resembles: in a **Z twist** the visible spirals run like the middle stroke of the letter Z (rising to the right); in an **S twist** they run like the middle of S. A rope's **lay direction** is quoted for the outer level — the strands in the finished rope. Most three-strand fibre rope is **Z-laid** (“right-laid”): the strands follow a Z helix, the strand twist is S, and the yarn twist within is Z — alternating at every level. Recording lay direction matters because a rope must be coiled and spliced with its lay, and because replacement rope in a system should match the original.

2.3 Lay length

The **lay length** is the axial distance for one strand to make one complete turn around the rope. A short lay (“hard lay”) gives a firm, abrasion-tolerant rope that holds its shape but is stiffer and loses more strength in the closing, because its elements run at a steep helix angle to the load. A long lay (“soft lay”) is more flexible and retains more axial strength, but opens more easily, snags, and distorts under handling. Lay length is therefore a specification, not an afterthought.

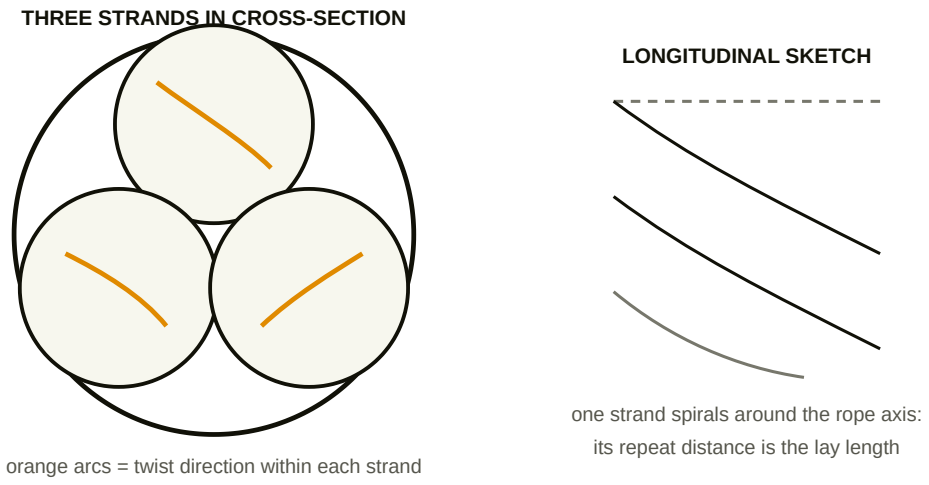


Figure 1 — Three-strand hawser-laid rope: cross-section showing the three strands arranged triangularly inside the rope outline (left), and a longitudinal sketch of the helical lay (right).

3. Braided and plaited constructions

- **Eight-strand plaited rope** — four pairs of strands interlace in a plait (two pairs S, two pairs Z), giving a square-ish rope that is torque-neutral: it does not tend to rotate under load, coils without kinking, and is widely used for mooring.
- **Double braid (braid-on-braid)** — a braided core carries much of the load inside a braided cover that protects it from abrasion and UV. Both parts share load, splicing is well standardised, and the rope is smooth and flexible — the standard construction for yacht sheets and many industrial ropes.
- **Single braid (hollow braid)** — one braided tube, often of HMPE, very light and strong, easily spliced, used for high-performance cordage; a separate cover may be added where abrasion is severe.
- **Kernmantle** — a twisted or braided core (kern) carries the load while a tightly braided sheath (mantle) protects it; the standard for climbing and rescue ropes, where the sheath also gives handling and knot behaviour.

Braided ropes are made on **braiding machines**: carriers holding strand packages travel in interweaving paths around the braiding point — half the carriers moving clockwise, half anticlockwise, passing in and out — while the finished rope is drawn away at a rate that sets the braid angle and picks per unit length.

4. Rope-making machinery in outline

- **Strand former** — combines the counted yarns, applies strand twist and tension, and winds the strand onto a large bobbin or former. Even tension between yarns is essential: a slack yarn carries no load until its neighbours have already stretched.
- **Rope closing machine** — closes strands into rope. In the **tubular** type the strand bobbins sit in a rotating tube and the strands are drawn through and laid as the tube rotates. In the **planetary** type the strand packages are carried around the rope axis on a rotating frame while being kept from rotating about their own axes, so the lay is applied without adding unwanted twist to the strands themselves. Either way, the finished rope passes through a closing die and onto a take-up reel under controlled tension.
- **Twisters for twine** — smaller two-for-one and ring-type twisters ply yarns into twines at high speed; the same S/Z discipline applies at this small scale.

Historical note for understanding, not nostalgia: the long **rope walk**, where yarns were stretched the full length of the rope before twisting, existed because twist and tension must be applied uniformly over the whole length at once — the modern machines are simply a way of doing the same thing in a rotating frame in a few metres of floor space.

5. Strength: why construction costs strength

A rope is always weaker than the sum of its yarns laid straight and parallel. The loss has identifiable causes:

- **Helix angle** — twisted elements pull partly sideways, not purely along the rope axis, so only the axial component of their strength is used.
- **Unequal load sharing** — elements at different radii and tensions do not reach breaking load simultaneously; the most loaded fail first and shed load onto the rest.
- **Bending and internal abrasion** — elements bend around each other at every crossover and rub during loading cycles.

The ratio actually achieved is the **construction (strength) efficiency**:

$$\text{Efficiency \%} = \text{rope breaking load} / \text{sum of yarn breaking loads} \times 100$$

Well-made laid and braided fibre ropes commonly achieve roughly **50–70%** — a standard textbook range, varying with fibre, lay and braid angle. Parallel-core constructions, in which the load-bearing elements run almost straight, reach higher efficiency, which is one reason they are used where maximum strength per weight matters.

WORKED EXAMPLE 1 — TWIST CONTRACTION ON CLOSING

Strands are consumed faster than rope is produced, because each strand follows a helix. Suppose 105 m of strand is consumed to produce 100 m of finished rope.

Contraction = (105 − 100) / 105 × 100 = 5 / 105 × 100 = **4.8%**.

Production planning must order strand length about 5% above rope length in this case — the same style of allowance as crimp in weaving (compare TE-3201, Topic 2).

WORKED EXAMPLE 2 — ESTIMATING ROPE BREAKING LOAD (ESTIMATE METHOD)

The yarns in a rope have a combined (summed) breaking load of 150 kN if tested straight and parallel. Take a construction efficiency of 60%.

Estimated rope breaking load = 150 × 0.60 = **90 kN**.

This is an **estimate method** for design teaching only: a rope offered for service must have its minimum breaking load established by testing actual rope specimens, never by this calculation alone.

6. Twines, ropes and cables compared

PRODUCT	CONSTRUCTION	TYPICAL SCALE	TYPICAL USES
Twine	2 or more yarns plied	Sold by length; ~1–3 mm	Baling, tying, fishing, packaging
Laid rope	3 (or 4) strands closed together	~4 mm upward	General purpose, mooring, lifting slings
Plaited / braided rope	8-strand plait, double braid, kernmantle	~4 mm upward	Mooring, yachting, climbing, rescue
Cable-laid rope	Three laid ropes laid together	Large diameters	Heavy marine duty
Steel wire rope	Steel-wire strands laid around a fibre or steel core	Wide range	Cranes, lifts, mining — high temperature and abrasion tolerance, but heavy and corrosion-prone without protection

Glossary

Hawser lay — the classic three-strand laid construction. **Closing** — twisting strands together to form rope. **Lay length** — axial distance of one complete strand turn. **S / Z twist** — direction of a helix, named after the central stroke of the letter it resembles. **Double braid** — braided core inside a braided cover. **Kernmantle** — load-bearing core protected by a braided sheath. **Construction efficiency** — rope breaking load as a percentage of the summed yarn breaking loads. **Contraction** — percentage by which strand length exceeds the rope length produced.

Key points

• Hierarchy: fibre → yarn → strand → rope; twist direction alternates at each level so the structure stays closed.
• Three-strand hawser lay is normally Z-laid; lay length trades firmness and abrasion behaviour against flexibility and retained strength.
• Plaited and braided ropes add torque neutrality, smooth handling and protective covers.
• Closing machines (tubular or planetary) and strand formers exist to apply uniform twist and tension over the whole length.
• Construction efficiency is typically about 50–70%: a rope never reaches the summed strength of its yarns.
• Rated breaking loads come from testing rope, not from the efficiency estimate.

Practice

1. Strand consumption for a rope order is 52.5 m of strand per 50.0 m of rope. Calculate the contraction percentage.
2. The summed breaking load of the yarns for a rope is again estimated with an efficiency of 60%, but the yarn sum is 200 kN. Estimate the rope breaking load, and state why this figure cannot be used as the rope's rated load.
3. Explain, with the alternating-direction rule, the twist directions of yarn, strand and rope in a standard Z-laid three-strand rope.
4. A harbour needs a mooring rope that will not rotate or kink when coiled on deck. Recommend a construction from this sheet and justify it against three-strand hawser lay.

Answer key

1. Contraction = (52.5 − 50.0) / 52.5 × 100 = 2.5 / 52.5 × 100 = **4.8%**.
2. Estimate = 200 × 0.60 = **120 kN**. It cannot be the rated load because efficiency is an assumed average, not a measured property of this rope: real load sharing, lay, fibre variation and splices all move the true value, so rating requires destructive tests on specimens of the actual rope.
3. Directions alternate at each level so each stage locks the previous one instead of untwisting it. For a Z-laid rope: the finished rope lay is **Z**, the strands are twisted **S**, and the yarns within the strands are twisted **Z**.
4. Recommend an **eight-strand plaited** rope (or double braid): its opposed S and Z strand pairs make it torque-neutral, so it does not spin under load or kink when coiled, unlike a laid rope which stores torque and can hockle. It also flakes/coils flat on deck. Accept its somewhat different splicing and inspection practice as the trade-off.